A Bayesian Approach to Lag Selection in Higher-Order Stationary Vector Autoregressions

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Table of Contents

Table of Contents

Motivation and Importance

- Vector Autoregressions are pivotal in economics for modeling multivariate time-series data, with key applications including:
  - Economic forecasting;
  - Shock identification.
- Proper model specification is critical:
  - **Too many lags:** Risks overfitting; poor generalization to unseen data.
  - **Too few lags:** Risks underfitting; missed dynamics in the data.

Lag Length Selection in VAR Models

- **Standard Practices in Economics:**
  - Lag length often chosen arbitrarily.
  - Based on intuition, data frequency, and trial and error.
- **Formalized Frequentist Approaches:**
  - Information Criterion, LRT for nested models, etc.
  - Still arbitrary; LRT sequence of comparisons and stopping rule.
  - Often lack of transparency, reducing reproducibility.
  - Cannot fully quantify uncertainty associated with lag selection.
- **Bayesian Approaches:**
  - Explicitly incorporate prior beliefs; useful in economics where domain expertise informs model specification.
  - Provide a full probabilistic description of uncertainty.

Bayesian Lag Selection Stationary VAR Models

- **Stationarity Assumption:**
  - Statistical properties of the time series remain invariant overtime.
  - Stationarity constrains the autoregressive coefficients to lie within the stationary region.
  - In multidimensional settings, stationary region becomes geometrically complex; prior specification becomes challenging.
- **Unconstrained Reparametrization (Heaps, 2023):**
  - Reparametrizes the original parameters to a set of unconstrained transformed partial autocorrelation matrices using a two-step bijective mapping.
  - Simplifies prior specification and allows for routine computational implementation since need only operate in the Euclidean Space.

Bayesian Lag Selection Stationary VAR Models

- **Bayesian Inference Framework (Binks et al., 2023):**
  - Introduces novel Bayesian inference framework for identifying the order of stationary VAR models.
  - Assigns a Multiplicative Gamma Process prior to the set of unconstrained transformed partial autocorrelations.
  - Promising results for low orders (e.g., one through four lags).
- **Still challenges with Higher Order VARs:**
  - Performance declines for moderately high lag orders, such as seven; consistent underestimation of the true order.
  - Theorization of excessive shrinkage from the original MGP prior specification.

My Contributions to this Framework

1. Introducing an extended MGP specification aiming at improving performance for higher-order VARs.
2. Designing a Monte Carlo algorithm to sample from the implied prior distribution over the effective lag order.
3. Conducting relevant simulation studies to compare the original and extended MGP specifications.

Table of Contents

Bayesian Inference for Lag Selection

- **Notation:**
  - P_s: s-th partial autocorrelation matrix,
  - A_s : s-th unconstrained matrix.
- **Role of shrinkage priors:**
  - By definition of partial autocorrelation matrices and the bijective mapping between P and A, a model of order k is nested within one of order $k+1$.
  - Shrinkage priors on $A_1, \ldots, A_{p_{\max}}$ (with large $p_{\max}$) identify the true lag order p by shrinking higher-order A_s matrices to zero.
- **Choice of prior:**
  - Common assumption that later terms (higher lags) should exert progressively less influence.
  - Multiplicative Gamma Process natural choice as it imposes increasing shrinkage on later lags.

Multiplicative Gamma Process (MGP)

The (i, j)th element in A_s is modeled hierarchically:

$$a_{s,ij} \mid \lambda_{s,ij}, \tau_s \sim \mathcal{N}\left(0, (\lambda_{s,ij}\tau_s)^{-1}\right),$$

independently for $i, j = 1, \ldots, m$ and $s = 1, \ldots, p_{\max}$.

The local precision parameters at lag s are assigned:

$$\lambda_{s,ij} \sim \text{Gamma}\left(\frac{a}{2}, \frac{a}{2}\right),$$

independently for all i, j, s.

Multiplicative Gamma Process (MGP)

The global shrinkage parameters at lag s are constructed as:

$$\tau_s = \prod_{k=1}^{s} \delta_k, \quad \delta_1 \sim \text{Gamma}(a_1, 1), \quad \delta_k \sim \text{Gamma}(a_2, 1) \text{ for } k \geq 2,$$

where δ_k are independent.

Global shrinkage depends on hyperparameter a_1 and a_2:

- $a_1 = 2.5$, $a_2 = 3$ (Binks et al., 2023).
- $a_1 = 2$, $a_2 = 2$: more flexible, accommodates more lags.

Introducing the Extended MGP

Proposed extension of the MGP imposes a prior Gamma distribution on the local shrinkage rates:

$$\lambda_{s,ij} \sim \text{Gamma}\left(\frac{a}{2}, \eta_{s,ij}\right), \quad \eta_{s,ij} \sim \text{Gamma}(c_1, c_2).$$

Key Points:

- Introduces flexibility; enables the model to adjust shrinkage rates across local precision parameters according to the data's specific characteristics.
- Hyperparameters c_1 and c_2 control the flexibility; if c_1 moderately larger than c_2 most draws smaller than one offsetting some of the global shrinkage.

Overall Prior Distribution

The overall prior distribution for the parameters in the model is specified as

$$\pi(\Sigma, A_1, \ldots, A_{p_{\max}}, \vartheta) = \pi(\Sigma)\, \pi(\vartheta) \prod_{s=1}^{p_{\max}} \pi(A_s \mid \vartheta),$$

where ϑ represent the collection of unknown hyperparameters in the Multiplicative Gamma Process.

Likelihood and Posterior Distribution

The likelihood function of a zero-mean vector autoregressive process of order $p_{\max}$ is given by:

$$p(y_{1:n} \mid \Sigma, \phi) = p(y_{1:p_{\max}} \mid \Sigma, \phi) \prod_{t=p_{\max}+1}^{n} p(y_t \mid y_{t-p_{\max}:t-1}, \Sigma, \phi).$$

Combining the prior distribution with the data through the Bayes rule gives rise to the posterior distribution:

$$\pi(\Sigma, A_1, \ldots, A_{p_{\max}}, \vartheta \mid y_{1:n}) \propto \pi(\Sigma)\pi(\vartheta) \prod_{s=1}^{p_{\max}} \pi(A_s \mid \vartheta) p(y_{1:n} \mid \Sigma, A_1, \ldots, A_{p_{\max}}).$$

Sampling: Since it is a complex function of the A_s, sampling through Hamiltonian Monte Carlo.

The Estimated Lag Order (Posterior Inference)

- The MGP prior does not assign probability mass at zero, so none of the A_s (P_s) values shrink to exactly zero.
- A truncation criterion determines when P_s is effectively zero.
- **Effective Lag Order (p^*):** The largest $s \leq p_{\max}$ such that P_s fails the truncation criterion, while all P_{s+k} for $k > 0$ satisfy it.

Prior Distribution p^* (Introducing the Monte Carlo Algorithm):

- The effective lag order is not directly parametrized; it emerges through shrinkage on A_s.
- A Monte Carlo algorithm was devised to sample from the induced prior distribution over the effective lag order.

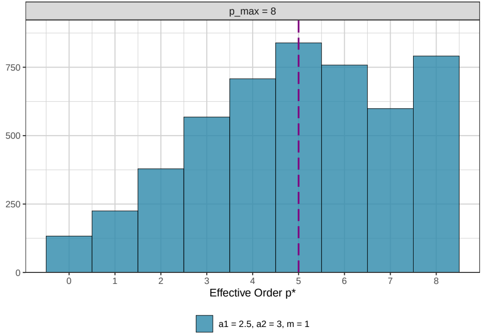
Simulation Design and Performance Evaluation

1. Compare the prior distribution over the effective lag order p^* induced by the original and extended MGP specifications across hyperparameter settings, dimension and overparametrization levels.
2. Compare the accuracy of the Bayesian inference framework with the original MGP specification versus the extended MGP specification in selecting the correct order for data generated from autoregressive processes of different orders.

Note: Some results for higher dimensions are available but are omitted as they closely match the presented findings.

Table of Contents

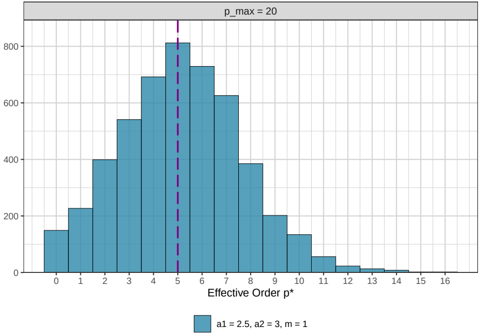
Induced p^* Prior Original MGP: $p_{\max} = 8$, $m = 1$



Note: These results were constructed using 5,000 draws of p^*.

- At low overparametrizations, the induced prior comfortably accommodates all considered lag orders.

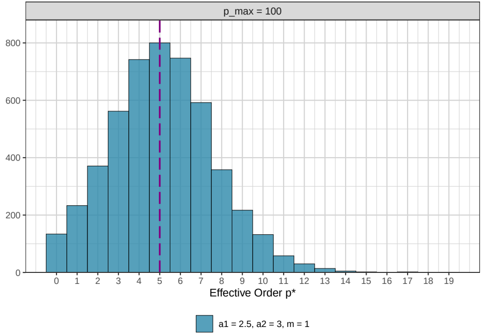
Induced p^* Prior Original MGP: $p_{\max} = 20$, $m = 1$



Note: These results were constructed using 5,000 draws of p^*.

- At moderate overparametrizations, the induced prior fails to assign meaningful mass beyond a certain order, preventing order identification in that range.

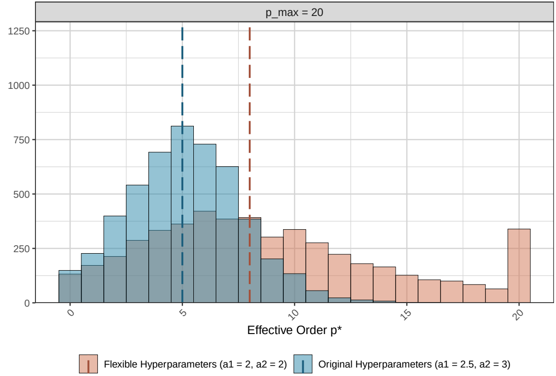
Induced p^* Prior Original MGP: $p_{\text{max}} = 100$, $m = 1$



Note: These results were constructed using 5,000 draws of p^*.

- Increasing the overparametrization has little effect on the induced prior distribution.

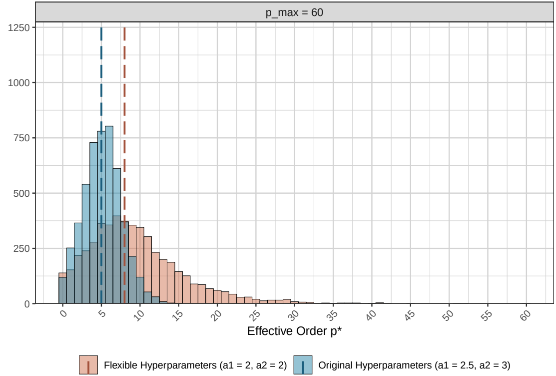
Induced p^* Priors Original MGP: $p_{\text{max}} = 20$, $m = 1$



Note: These results were constructed using 5,000 draws of p^*.

- With flexible hyperparameters, the original MGP induces a prior distribution that better accommodates higher lags.

Induced p^* Priors Original MGP: $p_{\text{max}} = 60$, $m = 1$

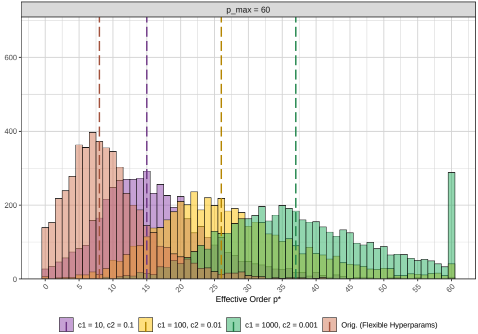


Note: These results were constructed using 5,000 draws of p^*.

- Even with flexible hyperparameter settings, excessive shrinkage persists for sufficiently large overparametrizations.

Table of Contents

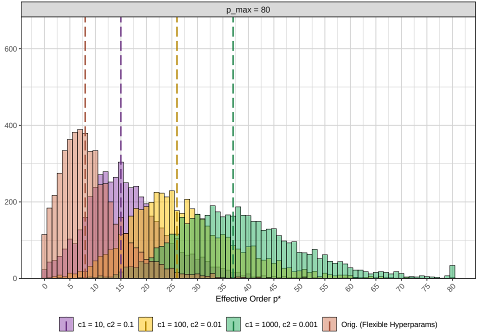
Induced p^* Priors Extended MGP: $p_{\max} = 60$, $m = 1$



Note: These results were constructed using 5,000 draws of p^*.

- Compared to the original MGP specification with flexible hyperparameters, the novel MGP specification induces a prior that accommodates higher lags more effectively.

Induced p^* Priors Extended MGP: $p_{\max} = 80$, $m = 1$

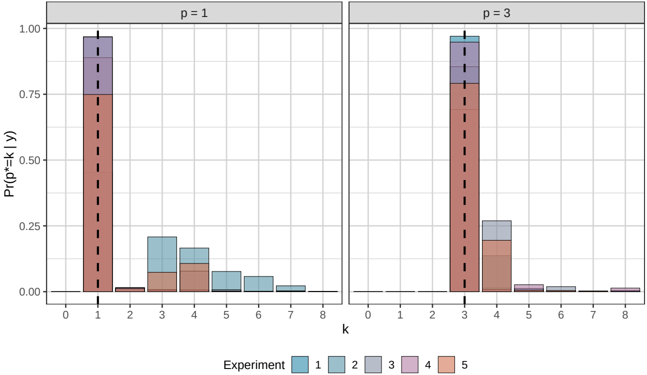


Note: These results were constructed using 5,000 draws of p^*.

- A wider gap between c_1 and c_2 enhances the prior distribution's ability to accommodate higher orders.

Table of Contents

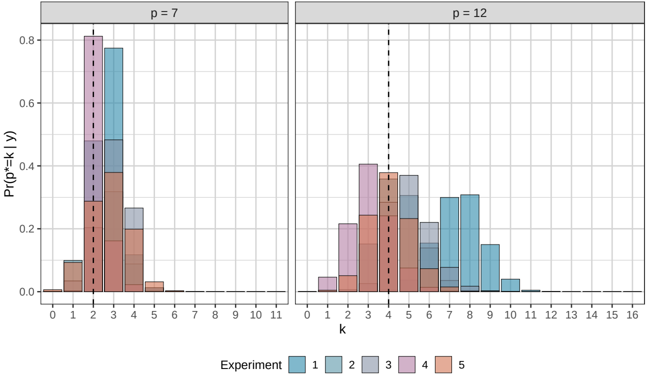
Overlayed Posterior p^* Original MGP: $p = 1, 3$; $m = 1$



Note: The simulated datasets contained 10,000 observations.

- **Low Orders:** The inference framework identifies the correct order when using the original MGP specification with original hyperparameters.

Overlayed Posterior p^* Original MGP: $p = 7, 12$; $m = 1$

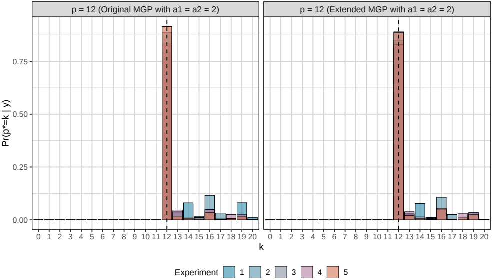


Note: The simulated datasets contained 10,000 observations.

- **Moderate orders:** Inference framework identifies incorrect order when using the original MGP specification with original hyperparameters.

Table of Contents

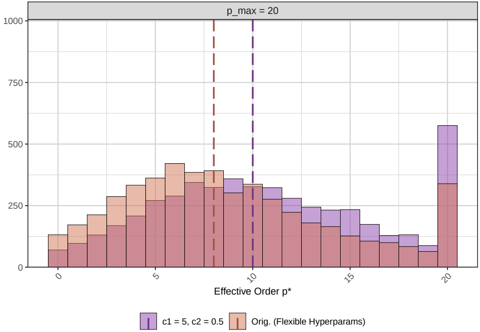
Overlayed Posterior p^* Extended MGP: $p = 12$, $m = 1$



Note: The simulated datasets contained 1,000 observations.

- Both the original MGP with flexible hyperparameters and the extended specification successfully identified the correct order of 12.

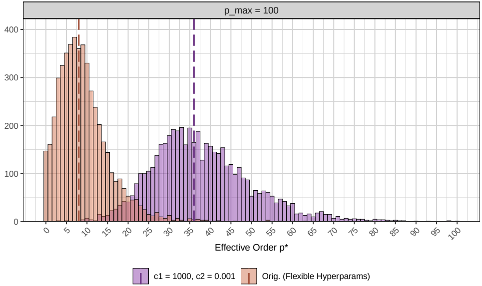
Overlayed Posterior p^* Extended MGP: $p = 12$, $m = 1$



Note: The simulated datasets contained 1,000 observations.

- This result is unsurprising, as the induced prior of both the original MGP specification with flexble hyperparameters and the extended specification accommodate 12 lags.

Overlayed Posterior p^* Extended MGP: $p = 12$, $m = 1$



Note: The simulated datasets contained 1,000 observations.

- For a process with $p = 80$, the original specification would fail to accommodate this order, as it falls outside the prior distribution's range. However, it would fall within the range of the novel specification under specific parameterizations.

Table of Contents

Limitations and Future Research

- Constructing a posterior for higher lag values could substantiate the novel MGP's advantage over the original with tuned hyperparameters.
- High computational costs made reliable results for large lag orders infeasible.
- Higher-lag scenarios offer a promising direction for validating the extended MGP in large-scale VARs.

Thank you!

Thank you for having me!

Table of Contents

Stationary VARs

A $VAR_m(p)$ represents an m-variate process $\{y_t\}$ with p lags:

$$y_t = \phi_1 y_{t-1} + \cdots + \phi_p y_{t-p} + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}_m(0, \Sigma),$$

where:

- $y_t \in \mathbb{R}^m$: vector of m variables at time t,
- $\phi_i \in \mathbb{R}^{m \times m}$: coefficient matrices $(i = 1, \ldots, p)$.

Stationarity Condition: Stationarity holds if and only if the roots of $det(I_m - \phi_1 z - \cdots - \phi_p z^p) = 0$ lie outside the unit circle.

Stationary Region: Subset of the parameter space where stationarity conditions are satisfied; grows increasingly complex with p and m making prior specification challenging.

Reparameterization (Heaps, 2023)

Idea: Reparameterization over the stationary region in terms of a set of interpretable, uncostrained parameters with a two-step bijective mapping:

1. Map $(\Sigma, \phi) \in S_m^+ \times \mathcal{C}_{p,m}$ to $\{\Sigma, P_1, \ldots, P_p\}$, where P_s is the $(s+1)$-step partial autocorrelation matrix:

$$P_{s+1} = \Sigma_s^{-1/2} \mathrm{Cov}(y_{t+1}, y_{t-s} \mid y_t, \ldots, y_{t-s+1}) \Sigma_s^{*\,-1/2}.$$

2. Map P_s to an unconstrained matrix $A \in \mathbb{R}^{m \times m}$ using:

$$A = (I_m - PP^\top)^{-1/2} P.$$

This allows for simplified prior specification and routine MCMC implementation since need only operate over a Euclidean space.

Truncation Criterion

The partial autocorrelation matrix at lag s is set to zero if the absolute value of all elements falls below a threshold:

$$\varepsilon = q_m(\beta) = \frac{1}{\sqrt{n}} \Phi^{-1} \left(\frac{\beta^{1/m^2} + 1}{2} \right),$$

where Φ^{-1} is the inverse CDF of the standard normal distribution:

- The lower the threshold, the harder it is to truncate lags.
- Threshold decreases with n (sample size).
- Threshold increases with β (probability parameter) and m (dimension).

Simulating Data from a Stationary VAR Process

1. Sample A_s elements independently from a standard normal distribution.
2. Transform A_s into partial autocorrelation matrices P_s using the bijective mapping, then P_s into the coefficient matrices ϕ using a recursive algorithm.
3. Ensure stationarity by verifying that all eigenvalues of the $mp \times mp$ coefficient matrix have magnitudes less than one.
4. Initialize the first p observations from a normal distribution.
5. Generate subsequent observations iteratively:

$$y_t = \mu + \sum_{i=1}^{p} \phi_i (y_{t-i} - \mu) + \epsilon_t,$$

where $\epsilon_t \sim \mathcal{N}(0, \Sigma)$.

Prior Distribution

By definition of partial autocorrelation matrices the order of a $\text{VAR}_m(p)$ model satisfies:

$$k < p \iff P_k \neq 0_m \quad \text{and} \quad P_{k+s} = 0_m \quad \text{for } s = 1, \ldots, p-k.$$

Given the bijective mapping between P and A

$$P_s = 0_m \quad \iff \quad A_s = 0_m.$$

Model Nesting: For $k < p$, the model is nested within that of $k+1$ under the unconstrained reparametrization.

Role of Shrinkage Priors: Shrinkage priors on $A_1, \ldots, A_{p_{\max}}$ (with large $p_{\max}$) allow identifying the true lag order p by shrinking higher-order A_s matrices to effectively zero.

Partial Autocorrelation Matrices

Each P_{s+1} represents the $(s+1)$-th partial autocorrelation matrix, defined as the conditional cross-covariance between y_{t+1} and y_{t-s}, given $y_t, \ldots, y_{t-s+1}$ which has been standardized through:

$$\Sigma_s^{-\frac{1}{2}} \operatorname{Cov}(y_{t+1}, y_{t-s} \mid y_t, \ldots, y_{t-s+1}) \, \Sigma_s^{* \, -\frac{1}{2}},$$

where:

- $\Sigma_s = \operatorname{Var}(y_{t+1} \mid y_t, \ldots, y_{t-s})$,
- $\Sigma_s^* = \operatorname{Var}(y_{t-s} \mid y_t, \ldots, y_{t-s+1})$,
- $\Sigma^{\frac{1}{2}}$: Symmetric matrix square root.

```
Input: samples, m, p_max, N, β, a_1, a_2, c_1, c_2, a, if_original
Output: optimal_p
optimal_p ← []
for i = 1 to samples do
    Generate τ (global shrinkage):
    τ[1] ← RGamma(a_1, 1)
    for s = 2 to p_max do
        τ[s] ← τ[s − 1] × RGamma(a_2, 1)
    end for
    Generate λ (local shrinkage) and A matrices:
    for s = 1 to p_max do
        for i = 1 to m do
            for j = 1 to m do
                if if_original then
                    η ← a
                else
                    η ← RGamma(c_1, c_2)
                end if
                λ[i, j, s] ← RGamma(a/2, η)
                variance ← 1 / (λ[i,j,s] × τ[s])
                A_matrices[i, j, s] ← RNormal(0, √variance)
            end for
        end for
    end for
```

    Convert A to P matrices:
    for $s = 1$ **to** $p_{\max}$ **do**
        $C \leftarrow \text{IdentityMatrix}(m) + A\_matrices[:,:,s] \times \text{Transpose}(A\_matrices[:,:,s])$
        $B \leftarrow \text{SqrtMatrix}(C)$
        $P\_matrices[:,:,s] \leftarrow \text{Inverse}(B) \times A\_matrices[:,:,s]$
    end for
    Generate threshold for truncation:
    $\text{threshold} \leftarrow \dfrac{\text{QNorm}\left(\left(\beta^{\frac{1}{m^2}} + 1\right)/2\right)}{\sqrt{N}}$
    Compute optimal p:
    $optimal\_p\_value \leftarrow 0$
    for $s = 1$ **to** $p_{\max}$ **do**
        if $\text{MaxAbsValue}(P\_matrices[:,:,s]) \leq \text{threshold}$ **then**
            break
        else
            $optimal\_p\_value \leftarrow s$
        end if
    end for
    $optimal\_p \leftarrow \text{Append}(optimal\_p\_value, optimal\_p)$
end for
Return: $optimal\_p$

Table: Hyperparameter settings for each scenario (Transposed)

Scenario	a1	a2	(c1, c2)
Original	2.5	3	-
Original (updated)	2	2	-
Extended	2	2	(10, 0.1), (100, 0.01), (1000, 0.001)

Note: $a = 6$, $\beta = 0.99$, samples = 5,000, $N = 10000$. VAR$_m(p_{max})$ processes include all combinations of $m = [1, 3, 5]$ and $p_{\max} = 10, 20, \ldots, 100$.

$c_1 = 5,\ c_2 = 0.5$					$c_1 = 10,\ c_2 = 0.1$				
$p_{\mathbf{max}}$	**Min**	**Max**	**Median**	**SD**	$p_{\mathbf{max}}$	**Min**	**Max**	**Median**	**SD**
$m = 1$					$m = 1$				
20	0	20	10	5.50	*40*	0	40	15	7.97
30	0	30	10	6.52	*50*	0	50	15	8.28
40	0	40	10	7.25	*60*	0	60	16	8.47